Qi Shi

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Academic Appointments

Kent State University, Kent, OH, USA **Sept. 2025 - present**
Postdoctoral Scholar in the Physics Department

Brookhaven National Laboratory, Upton, NY, USA **Sept. 2022 - Sept. 2025**
Visitor in the Physics Department

Education

Central China Normal University, Wuhan, Hubei, China **Sept. 2021 - Jun. 2025**
Ph.D. in Physics

Central China Normal University, Wuhan, Hubei, China **Sept. 2018 - Jun. 2021**
M.S. in Physics

Northeast Normal University, Changchun, Jilin, China **Sept. 2014 - Jun. 2018**
B.S. in Physics

Publications (INSPIRE Author Profile)

Journal Articles (* denotes corresponding author)

[1] Xiang Gao, Andrew Hanlon, Jin-Chen He, Xiang-Yu Jiang, Swagato Mukherjee, Peter Petreczky, **Qi Shi***, and Yong Zhao. "Nucleon Unpolarized PDFs from Lattice QCD at Physical Point with High Boost". ***In preparation*** (2026).

[2] Xiang Gao, Swagato Mukherjee, **Qi Shi***, Fei Yao*, and Yong Zhao. "Skewness-dependent moments of the pion GPD from nonlocal quark-bilinear correlators". ***Phys. Rev. D*** 113.1 (2026), 014505. DOI: 10.1103/hrb2-47zm. arXiv: 2511.01818 [hep-lat].

[3] Heng-Tong Ding, Xiang Gao, Swagato Mukherjee, Peter Petreczky, **Qi Shi***, Sergey Syritsyn, and Yong Zhao. "Three-dimensional imaging of pion using lattice QCD: generalized parton distributions". ***JHEP*** 02 (2025), 056. DOI: 10.1007/JHEP02(2025)056. arXiv: 2407.03516 [hep-lat].

[4] Heng-Tong Ding, Xiang Gao*, Andrew Hanlon, Swagato Mukherjee, Peter Petreczky, **Qi Shi***, Sergey Syritsyn, Rui Zhang, and Yong Zhao. "QCD Predictions for Meson Electromagnetic Form Factors at High Momenta: Testing Factorization in Exclusive Processes". ***Phys. Rev. Lett.*** 133 (18 Oct. 2024), 181902. DOI: 10.1103/PhysRevLett.133.181902. arXiv: 2404.04412 [hep-lat]. (Reported by BNL News, BNL's Top 10, ScienceDaily, Phys.org, and EurekAlert).

[5] Xiang Gao*, Andrew Hanlon*, Swagato Mukherjee, Peter Petreczky, **Qi Shi**, Sergey Syritsyn, and Yong Zhao. "Transversity PDFs of the proton from lattice QCD with physical quark masses". ***Phys. Rev. D*** 109.5 (2024), 054506. DOI: 10.1103/PhysRevD.109.054506. arXiv: 2310.19047 [hep-lat].

[6] Heng-Tong Ding*, Sheng-Tai Li, **Qi Shi**, and Xiao-Dan Wang. "Fluctuations and correlations of net baryon number, electric charge and strangeness in a background magnetic field". ***Eur. Phys. J. A*** 57.6 (2021), 202. DOI: 10.1140/epja/s10050-021-00519-3. arXiv: 2104.06843 [hep-lat].

Conference Proceedings (* denotes speaker)

[1] Heng-Tong Ding*, Xiang Gao, Andrew Hanlon, Swagato Mukherjee, Peter Petreczky, Qi Shi, Sergey Syritsyn, Rui Zhang, and Yong Zhao. "Probing Meson Structure via Lattice QCD: Electromagnetic Form Factor at High Momentum Transfer and Generalized Parton Distribution". *PoS* SPIN2025 (2026), 086. DOI: 10.22323/1.517.0086.

[2] Heng-Tong Ding, Xiang Gao, Swagato Mukherjee, Peter Petreczky, **Qi Shi***, Sergey Syritsyn, and Yong Zhao. "Lattice QCD calculation of the pion generalized parton distribution". *PoS* SPIN2023 (2024), 024. DOI: 10.22323/1.456.0024.

[3] Heng-Tong Ding, Xiang Gao, Andrew Hanlon, Swagato Mukherjee, Peter Petreczky, **Qi Shi***, Sergey Syritsyn, and Yong Zhao. "Lattice QCD predictions of pion and kaon electromagnetic form factors at large momentum transfer". *PoS* LATTICE2023 (2024), 320. DOI: 10.22323/1.453.0320.

[4] Heng-Tong Ding*, Sheng-Tai Li, **Qi Shi**, Akio Tomiya, Xiao-Dan Wang, and Yu Zhang. "QCD phase structure in strong magnetic fields". *Acta Phys. Polon. Supp.* 14 (2021), 403. DOI: 10.5506/APHYSPOLBSUPP.14.403. arXiv: 2011.04870 [hep-lat].

Computing Resource Awards (as Principal Investigator)

1. Jan. 2026 - Jan. 2027: NERSC Allocation Year 2026 DOE Mission Science Allocation Award - "Nucleon Generalized Parton Distributions from lattice QCD at Physical Masses", 63,000 GPU node hours.

2. Jan. 2025 - Jan. 2026: NERSC Allocation Year 2025 DOE Mission Science Allocation Award - "Computing Pion Generalized Parton Distributions at Non-zero Skewness", 32,000 GPU node hours.

Oral Presentations

1. "Nucleon Unpolarized PDFs from Lattice QCD at Physical Point with High Boost", The 43rd International Symposium on Lattice Field Theory **(Lattice 2026)**, July 26 - August 1, 2026, University of Maryland, USA.

2. "Nucleon Unpolarized PDFs at the Physical Point in the Coulomb Gauge" (online talk), 2026 Meeting on Lattice Parton Physics from Large Momentum Effective Theory **(LaMET 2026)**, July 6 - 9, 2026, Jagiellonian University, Poland.

3. "Exploring Meson Structure via Lattice QCD" (online talk), **Invited seminar talk**, High Energy Nuclear Physics in China, April 2, 2026.

4. "Exploring Meson Structure with Lattice QCD", **Invited CNR seminar talk**, November 4, 2025, Kent State University, USA.

5. "Exploring Meson Structure with Lattice QCD", **Invited RBRC seminar talk**, June 26, 2025, Brookhaven National Laboratory, USA.

6. "Exploring Meson Structures with Lattice QCD: Form Factors at High Momentum Transfer and Generalized Parton Distributions at Zero Skewness", **Invited talk**, Virtual Lattice Field Theory Colloquium at MIT, March 6, 2025.

7. "Exploring Meson Structures from Lattice: FF at large momentum transfer and GPD", Parton Distributions and Lattice Calculations **(PDFLattice 2024)**, November 18 – 20, 2024, Thomas Jefferson National Accelerator Facility, USA.

8. "Three-dimensional Imaging of Pion using Lattice QCD: Generalized Parton Distributions", 2024 Meeting on Lattice Parton Physics from Large Momentum Effective Theory **(LaMET2024)**, August 11 - 14, 2024, University of Maryland, USA.

9. "Exploring Meson Structures from Lattice QCD", From Quarks and Gluons to the Internal Dynamics of Hadrons **(CFNS workshop)**, May 15 - 17, 2024, Stony Brook University, USA.

10. "Lattice QCD Calculation of the Pion Generalized Parton Distributions (GPDs)", Workshop on Generalized Parton Distributions for Nucleon Tomography in the EIC Era **(RBRC workshop)**, January 17 – 19, 2024, Brookhaven National Laboratory, USA.

11. “Lattice QCD Calculation of the Pion Generalized Parton Distributions (GPDs)”, 25th International Spin Symposium **(SPIN 2023)**, September 24 - 29, 2023, Durham Convention Center, USA.
12. “Lattice QCD Prediction of Pion and Kaon Form Factor at Large Momentum Transfer Q^2”, **Invited seminar talk**, August 29, 2023, Argonne National Laboratory, USA.
13. “Lattice QCD Prediction of Pion and Kaon Form Factor at Large Momentum Transfer Q^2”, The 40th International Symposium on Lattice Field Theory **(Lattice 2023)**, July 31 - August 4, 2023, Fermi National Accelerator Laboratory, USA.
14. “Transversity PDFs of the Proton From Lattice QCD with Physical Quark Masses” (online talk), 2023 Meeting on Lattice Parton Physics from Large Momentum Effective Theory **(LaMET 2023)**, July 24 - 26, 2023, University of Regensburg, Germany.
15. “Lattice QCD Prediction of Pion and Kaon Form Factor at Large Momentum Transfer Q^2”, 2022 Meeting on Lattice Parton Physics from Large-Momentum Effective Theory **(LaMET 2022)**, December 1 – 3, 2022, Argonne National Laboratory, USA.
16. “Lattice QCD Prediction of Kaon Form Factor at Large Momentum Transfer up to 10 GeV^2” (online talk), American Physical Society April Meeting 2022 **(APS 2022)**, April 9 - 12, 2022.

Skills

High-Performance Computing	: NERSC, DELTA, SDCC, NSC3
Job Scheduling	: Slurm
Lattice QCD Software	: Qlua (Qlu+CUDA), Grid
Programming & Scripting	: Python, Bash
Data Analysis & Visualization Tools	: numpy, scipy, sympy, pandas, matplotlib, Wolfram Mathematica, WebPlotDigitizer
Development Environments & Editors	: VS Code, Jupyter Notebook, Vim
Documentation & Typesetting	: LaTeX
Languages	: Chinese (Native), English (Fluent)